UNIVERSITY OF MAKATI

College of Computing and Information Sciences

CHAPTER 3 REPORT: MARKET SIZING & PRODUCT COSTING

Technopreneurship • Academic Year 2026–2027

Project Title:	**HULOG: IoT-Enabled Dynamic QR Parcel Drop Vault**
Proponents:	CAMARADOR, Jhorenz S. • CORPUZ, Lance Josh M. • GOZE, AJ M. • LUZANO, Justin Mark D.
Course / Year:	BSIT — 4th Year (IV-BINS) • Technopreneurship
Professor / Adviser:	Engr. Bhai Nhuraisha I. Deplomo, PECE, MSEcE, MSCIT

1. REVIEW OF CHAPTER 3: TAM, SAM, AND SOM FRAMEWORK

Chapter 3 provides the foundational methodologies for identifying market opportunities and sizing target markets. In technopreneurship, market sizing allows founders and investors to understand the scale of a problem, determine commercial feasibility, and avoid pursuing an innovation that has no paying audience. The framework breaks down the potential market into three distinct, realistic layers:

- **TAM (Total Addressable Market):** The maximum theoretical demand for a product or service if 100% market share were captured nationwide. It represents the total population of individuals or businesses that experience the problem and could benefit from the solution, assuming no competition and zero distribution barriers.
- **SAM (Serviceable Addressable Market):** The specific subset of the TAM that our product can realistically serve based on our chosen business model, physical product design, technical constraints, geographic coverage, and customer income level. It eliminates demographics that do not match our channel or form factor.
- **SOM (Serviceable Obtainable Market):** The practical, realistic portion of the SAM that our team can capture during the first 1 to 3 years of commercial operations. This is directly constrained by early production limits, working capital, student sales bandwidth, and initial target beachheads.

Furthermore, Chapter 3 emphasizes that market sizing cannot be calculated in a vacuum—it requires an accurate **Target Selling Price**. Deriving an accurate price demands bottom-up expense accounting, where direct materials, labor/manpower, and production overhead are itemized to determine the unit Cost of Goods Sold (COGS).

2. PROJECT OVERVIEW & PROBLEM STATEMENT

Hulog is an autonomous, smart parcel drop vault installed at residential perimeter gates, townhouses, and student dormitories.

Over 65% of e-commerce orders in the Philippines (Shopee, Lazada, TikTok Shop) are prepaid through e-wallets such as GCash and Maya. However, because working homeowners and students are frequently away during daytime courier delivery windows, couriers often throw packages over gates or leave them exposed on doorsteps. This leads to rampant parcel theft, weather damage, or delivery rescheduling. Traditional padlocks require physical keys that cannot be shared remotely, while static QR stickers on mailboxes are easily photocopied or cloned.

Hulog solves this by using dynamic, single-use access credentials. When a parcel is scheduled, the owner generates a 6-digit PIN or dynamic QR code valid for 15 to 30 minutes. The courier punches the PIN on a

weatherproof 4x4 keypad with zero mobile app downloads. An ESP32 microcontroller triggers a 12V electromagnetic solenoid lock, an internal optical infrared (IR) sensor verifies parcel placement, a door switch confirms closure, and the single-use token immediately invalidates while sending an instant confirmation to the recipient.

3. PRODUCT EXPENSE & COST COMPUTATION (UNIT COGS)

To calculate the unit cost of a commercial finished vault, production expenses are divided into direct materials, direct manpower, and manufacturing overhead based on batch manufacturing runs.

A. Direct Material Costs (Bill of Materials)

Component / Subsystem	Engineering Description & Function	Cost (PHP)
ESP32 DevKit V1	Dual-Core 240MHz, Wi-Fi/BLE controller running FreeRTOS	₱240.00
12V Solenoid Latch Lock	12V 0.8A heavy-duty cabinet electromagnetic tongue latch	₱190.00
12V 2A DC Power Adapter	220V AC to 12V DC power supply adapter	₱115.00
0.96" I2C OLED Display	SSD1306 monochrome screen for dynamic QR and PIN prompts	₱110.00
4x4 Matrix Membrane Keypad	16-key weather-resistant keypad for courier PIN entry	₱65.00
LM2596 Buck Converter	Steps 12V down to 5V to power ESP32 from a single power line	₱55.00
MC-38 Magnetic Door Switch	Normally closed reed switch to verify door closure	₱55.00
1-Channel 5V Relay Module	Optocoupler isolation to switch 12V solenoid from ESP32 GPIO	₱45.00
Optical IR Sensor Module	Infrared sensor to detect physical parcel presence inside vault	₱25.00
5V Active Buzzer	Acoustic chime on unlock and alarm if door is held open	₱20.00
Wiring Harness & Terminals	Internal jumpers, terminal screw blocks, and DC jacks	₱80.00
Sheet Metal Enclosure	1.2mm Galvanized Iron (GI) steel box with anti-fishing baffle	₱500.00
Outdoor Powder Coating	Anti-rust, weather-resistant electrostatic matte finish	₱120.00
Hardware Fittings	Stainless piano hinge, rubber weather gaskets, security screws	₱80.00
Subtotal — Direct Materials (BOM):		**₱1,700.00**

B. Direct Manpower • Labor Cost

Calculated based on standard Philippine electronics assembly rates of ₱90.00 per hour (2.0 hours total labor per unit):

- **Wiring and Soldering (45 mins):** Wire harness crimping, board soldering, buck converter calibration = **₱67.50**
- **Mechanical Assembly (45 mins):** Solenoid alignment, keypad mounting, sensor installation = **₱67.50**

- **Firmware & QA Testing (30 mins):** ESP32 flashing, Wi-Fi pairing test, 10-cycle door lock inspection = **₱45.00**

Subtotal — Direct Manpower: ₱180.00

C. Manufacturing Overhead & Production Expenses

- **Packaging & Cushioning:** Corrugated box, custom foam inserts, user quick-start manual = **₱65.00**
- **Assembly Consumables:** Heat shrink, solder lead, cable ties, acrylic sealant = **₱25.00**
- **Shop Electricity & Tool Depreciation:** Power tools, soldering equipment allocation = **₱30.00**

Subtotal — Manufacturing Overhead: ₱120.00

TOTAL UNIT COST OF GOODS SOLD (COGS):	**₱2,000.00**

4. TARGET PRICE DETERMINATION & MARGINS

Our pricing strategy is benchmarked against non-smart mechanical drop boxes (₱2,500 to ₱4,500) and commercial digital door locks (₱4,000 to ₱8,000). A base selling price of **₱3,499.00** establishes an accessible entry point while maintaining a healthy gross margin:

Commercial Product Offering	Unit COGS	Selling Price	Gross Profit	Gross Margin
Hulog Core (Standard Turnkey Unit)	₱2,000.00	**₱3,499.00**	₱1,499.00	42.8%
Hulog Pro (w/ Li-Ion Battery Backup)	₱2,450.00	**₱4,499.00**	₱2,049.00	45.5%
Multi-Tenant SaaS (Monthly Subscription)	₱25.00/mo	**₱199.00/mo**	₱174.00/mo	87.4%

Customer Value Anchor: Preventing the theft or weather destruction of just one mid-range parcel (sneakers, cosmetics, or smartphones averaging ₱3,000 to ₱15,000) completely pays for the entire lifetime cost of the vault.

5. APPLICATION OF MARKET SIZING (TAM • SAM • SOM)

Applying Chapter 3 formulas using Philippine Statistics Authority (PSA) demographic data and local e-commerce delivery indices:

A. Total Addressable Market (TAM)

`TAM = 11,000,000 households × ₱3,500 = ₱38,500,000,000 (₱38.5 Billion)`

The Philippines has approximately 26.4 million households. According to e-commerce industry reports, approximately 42% (11.0 million households) regularly purchase online via Shopee, Lazada, and TikTok Shop at least once a month. At our base retail price of ₱3,500, the total nationwide opportunity is ₱38.5 Billion (~$687M USD).

B. Serviceable Addressable Market (SAM)

`SAM = 1,500,000 target units × ₱3,500 = ₱5,250,000,000 (₱5.25 Billion)`

We filter the TAM to the demographic segment that matches Hulog's physical gate form factor and distribution reach:

- *Geography:* Mega Manila (NCR and Region IV-A CALABARZON), which represents over 65% of all e-commerce parcel deliveries in the country.
- *Dwelling Type:* Standalone homes with perimeter gates, gated townhouses, and multi-tenant boarding houses (excluding high-rise condominium units that have private lobby mailrooms).
- *Income & Connectivity:* Middle-to-upper-middle-class households (SEC B and C) with active home Wi-Fi and working occupants who are regularly away during daytime delivery hours.

This yields an addressable market of 1.5 million target dwellings, representing a SAM of ₱5.25 Billion.

C. Serviceable Obtainable Market (SOM)

`SOM = 10,000 units × ₱3,500 = ₱35,000,000 (₱35.0 Million)`

Our conservative, realistic 3-year commercial target captures approximately **0.65% market share of our SAM** (10,000 cumulative units sold):

- *Year 1 (1,000 units • ₱3.5M):* Beachhead rollout to student dormitories and boarding houses around the University Belt (Manila, Makati, Quezon City) where communal gates receive dozens of daily parcels.
- *Year 2 (3,500 units • ₱12.25M):* Online direct-to-consumer store (Shopee/Lazada) and bulk partnerships with Homeowners Associations (HOAs) in Cavite, Rizal, and Taguig.
- *Year 3 (5,500 units • ₱19.25M):* Expansion into hardware retail distribution networks (e.g., Ace Hardware, Wilcon) across Metro Manila and Metro Cebu.

Adding recurring SaaS revenue from 1,200 subscribing boarding houses by Year 3 (1,200 × ₱199 × 12 months = ₱2.86M/year), our total 3-year obtainable revenue is approximately **₱37.85 Million**.

6. CONCLUSION & FEASIBILITY SUMMARY

The financial calculations demonstrate that **Project Hulog is commercially and technically feasible**. By engineering the electronics BOM to ₱1,000.00 and complete unit COGS to ₱2,000.00, our target price of ₱3,499.00 yields a strong gross margin of 42.8% without outpricing middle-class Filipino households. Capturing even less than 1% of the serviceable market in Mega Manila validates that Hulog represents a viable, sustainable business venture.